3.5.1 Additional homework ¶

Homework 3.5.1.1. Consider the matrix $\left(\begin{array}{c} A \\ \hline B \end{array}\right)$ where A has linearly independent columns. Let

- $A = Q_A R_A$ be the QR factorization of A.
- $\left(\begin{array}{c} R_A \\ \hline B \end{array}\right) = Q_B R_B$ be the QR factorization of $\left(\begin{array}{c} R_A \\ \hline B \end{array}\right)$.
- $\left(\begin{array}{c} A \\ \hline B \end{array}\right) = QR$ be the QR factorization of $\left(\begin{array}{c} A \\ \hline B \end{array}\right)$.

Assume that the diagonal entries of R_A, R_B, and R are all positive. Show that $R = R_B$.

▾ Solution

$$\left(\begin{array}{c} A \\ \hline B \end{array}\right) = \left(\begin{array}{c} Q_A R_A \\ \hline B \end{array}\right) = \left(\begin{array}{c|c} Q_A & 0 \\ \hline 0 & I \end{array}\right) \left(\begin{array}{c} R_A \\ \hline B \end{array}\right) = \left(\begin{array}{c|c} Q_A & 0 \\ \hline 0 & I \end{array}\right) Q_B R_B$$

Also, $\left(\begin{array}{c} A \\ \hline B \end{array}\right) = QR$. By the uniqueness of the QR factorization (when the diagonal elements of the triangular matrix are restricted to be positive), $Q = \left(\begin{array}{c|c} Q_A & 0 \\ \hline 0 & I \end{array}\right) Q_B$ and $R = R_B$.

Remark 3.5.1.1. This last exercise gives a key insight that is explored in the paper

- [21] Brian C. Gunter, Robert A. van de Geijn, Parallel out-of-core computation and updating of the QR factorization, ACM Transactions on Mathematical Software (TOMS), 2005.